

# Statistical Machine Learning (BE4M33SSU)

## Empirical Risk Minimization

Czech Technical University in Prague  
V. Franc

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# Learning

- ◆ **Goal:** Given a training set  $\mathcal{T}^m \sim p^m$ , find a strategy  $h: \mathcal{X} \rightarrow \mathcal{Y}$  with minimizing the generalization error  $R(h) = \mathbb{E}_{(x,y) \sim p}[\ell(y, h(x))]$ .
- ◆ **Hypothesis class (space):** is fixed before learning based on prior knowledge

$$\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}} = \{h: \mathcal{X} \rightarrow \mathcal{Y}\}$$

- ◆ **Learning algorithm:** is a function

$$A: \cup_{m=1}^{\infty} (\mathcal{X} \times \mathcal{Y})^m \rightarrow \mathcal{H}$$

returns a strategy  $h_m = A(\mathcal{T}^m)$  from  $\mathcal{H}$  based on a training set  $\mathcal{T}^m$

# Empirical Risk Minimization approach to learning

- ◆ The generalization error  $R(h)$  is approximated by the empirical risk  $R_{\mathcal{T}^m}(h)$  computed on the training examples  $\mathcal{T}^m \sim p^m$ :

$$R_{\mathcal{T}^m}(h) = \frac{1}{m}(\ell(y^1, h(x^1)) + \dots + \ell(y^m, h(x^m))) = \frac{1}{m} \sum_{i=1}^m \ell(y^i, h(x^i))$$

- ◆ The ERM based learning algorithm returns  $h_m$  such that

$$h_m \in \underset{h \in \mathcal{H}}{\operatorname{Argmin}} R_{\mathcal{T}^m}(h)$$

- ◆ Depending on the choice of  $\mathcal{H}$  and  $\ell$  and algorithm solving (1) we get individual instances e.g. Support Vector Machines, Linear Regression, Logistic Regression, Neural Networks learned by back-propagation, AdaBoost, Gradient Boosted Trees, ...

## ERM can fail due to overfitting

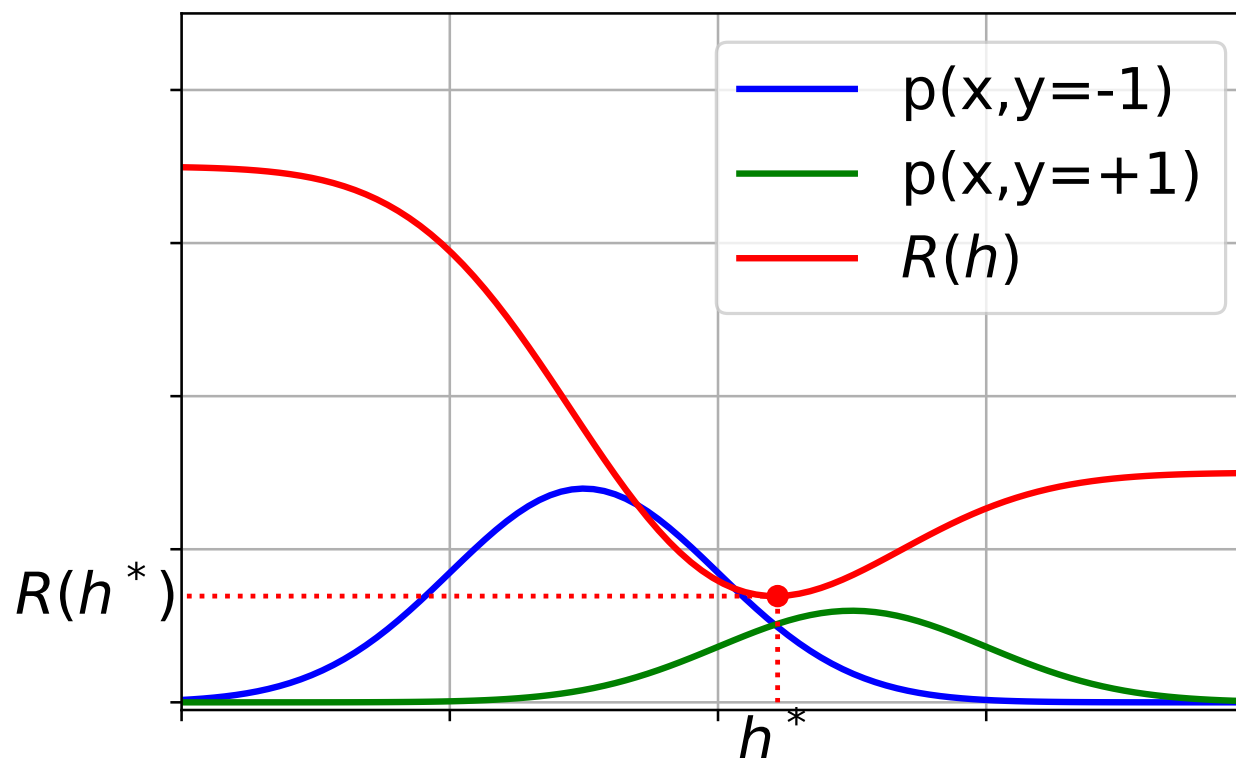
- ◆ Let  $\mathcal{X} = [a, b] \subset \mathbb{R}$ ,  $\mathcal{Y} = \{+1, -1\}$ ,  $\ell(y, y') = \mathbb{I}[y \neq y']$ ,  $p(x \mid y = +1)$  and  $p(x \mid y = -1)$  be uniform distributions on  $\mathcal{X}$  and  $p(y = +1) = 0.8$ .
- ◆ The optimal strategy is  $h(x) = +1$  with the Bayes risk  $R^* = 0.2$ .
- ◆ Learning algorithm “lookup table”: given training set  $\mathcal{T}^m$  it returns

$$h_m(x) = \begin{cases} y^j & \text{if } x = x^j \text{ for some } j \in \{1, \dots, m\} \\ -1 & \text{otherwise} \end{cases}$$

- Implements ERM principle as  $\mathbb{P}(R_{\mathcal{T}^m}(h_m) = 0) = 1$ .
- Fails to find a good solution as  $\mathbb{P}(R(h_m) = 0.8) = 1, \forall m \in \mathbb{N}$ .
- ◆ **Overfitting:** the case when  $h_m = A(\mathcal{T}^m)$  has a low empirical risk (training error)  $R_{\mathcal{T}^m}(h_m)$  while the expected risk  $R(h_m)$  is high.
- ◆ **Conclusion:** The empirical risk  $R_{\mathcal{T}^m}(h_m)$  is not in general a reliable proxy of the expected risk  $R(h)$ . In this lecture, we demonstrate that imposing constraints on the hypothesis space  $\mathcal{H}$  resolves this issue.

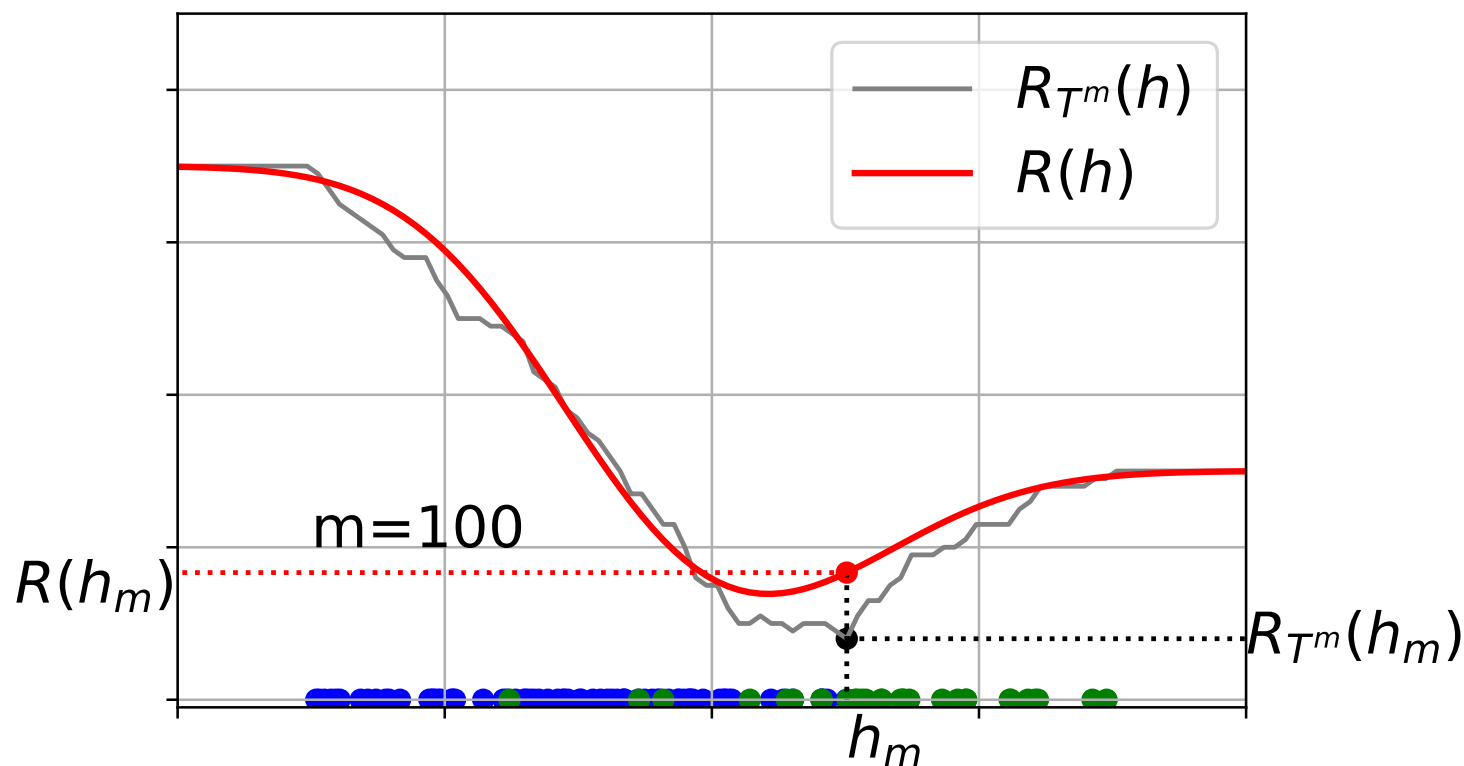
# For learning Law of Large Numbers is not enough we need Uniform Law of Large Numbers

$$\ell(y, y') = \mathbb{I}[y \neq y'], R(h) = \mathbb{E}_{(x,y) \sim p}[\ell(y, h(x))], h^* \in \operatorname{Argmin}_{h \in \mathcal{H}} R(h)$$



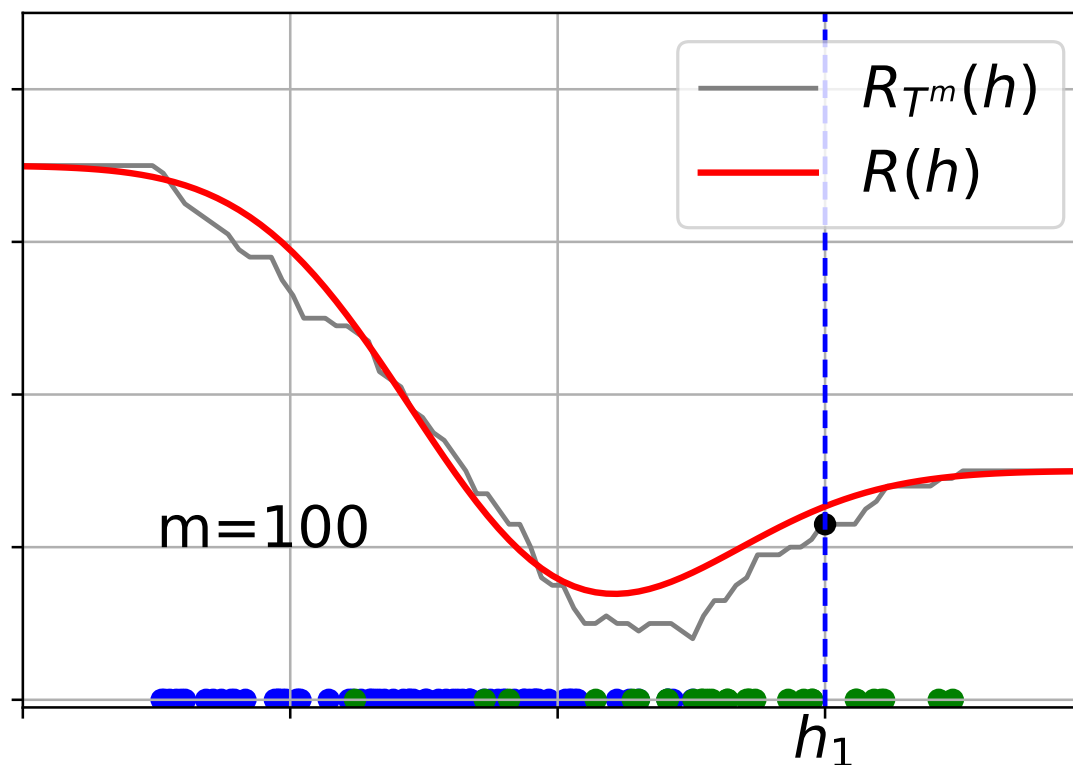
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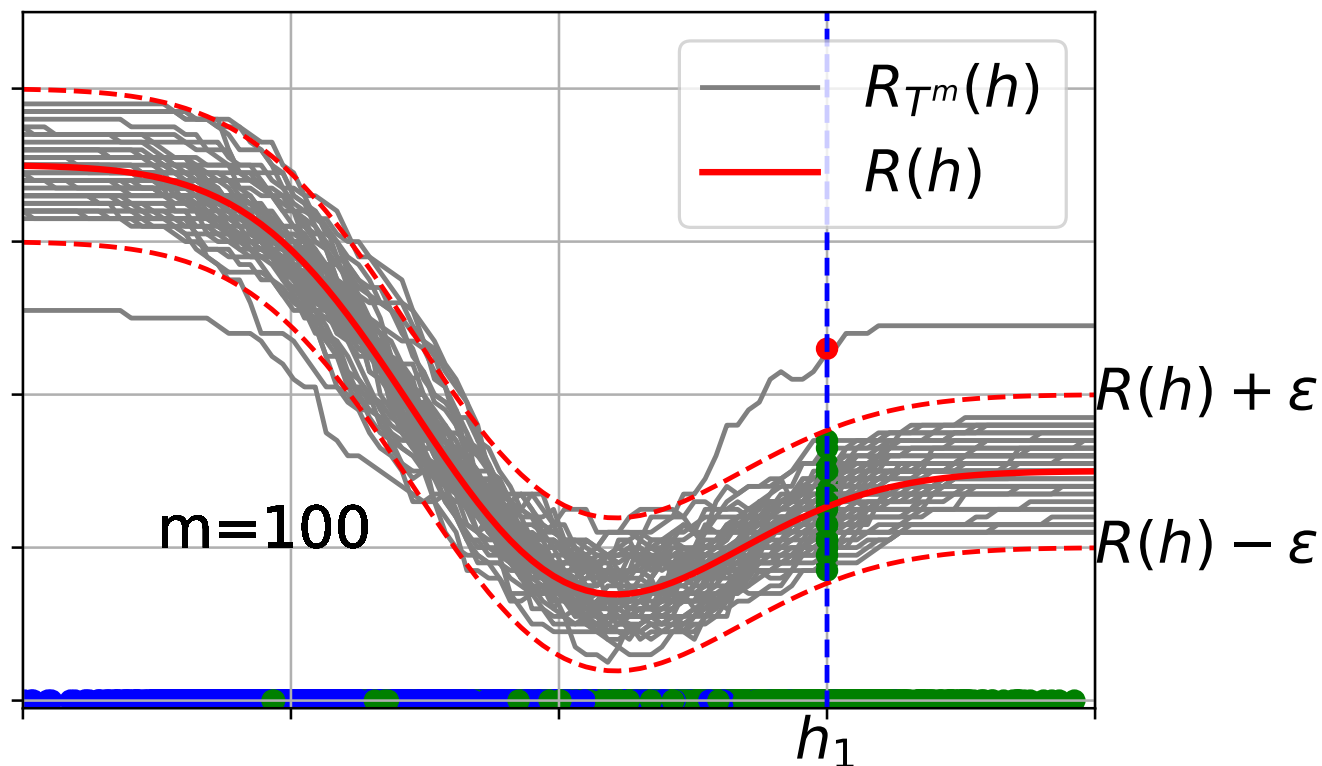
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LLN:  $l \geq \frac{\log(2) - \log(\delta)}{2\varepsilon^2} (\ell_{\max} - \ell_{\min})^2 \Rightarrow \mathbb{P}_{\mathcal{S}^l \sim p^l} \left( \underbrace{|R(h) - R_{\mathcal{S}^l}(h)|}_{\text{ER fails}} \geq \varepsilon \right) \leq \delta$

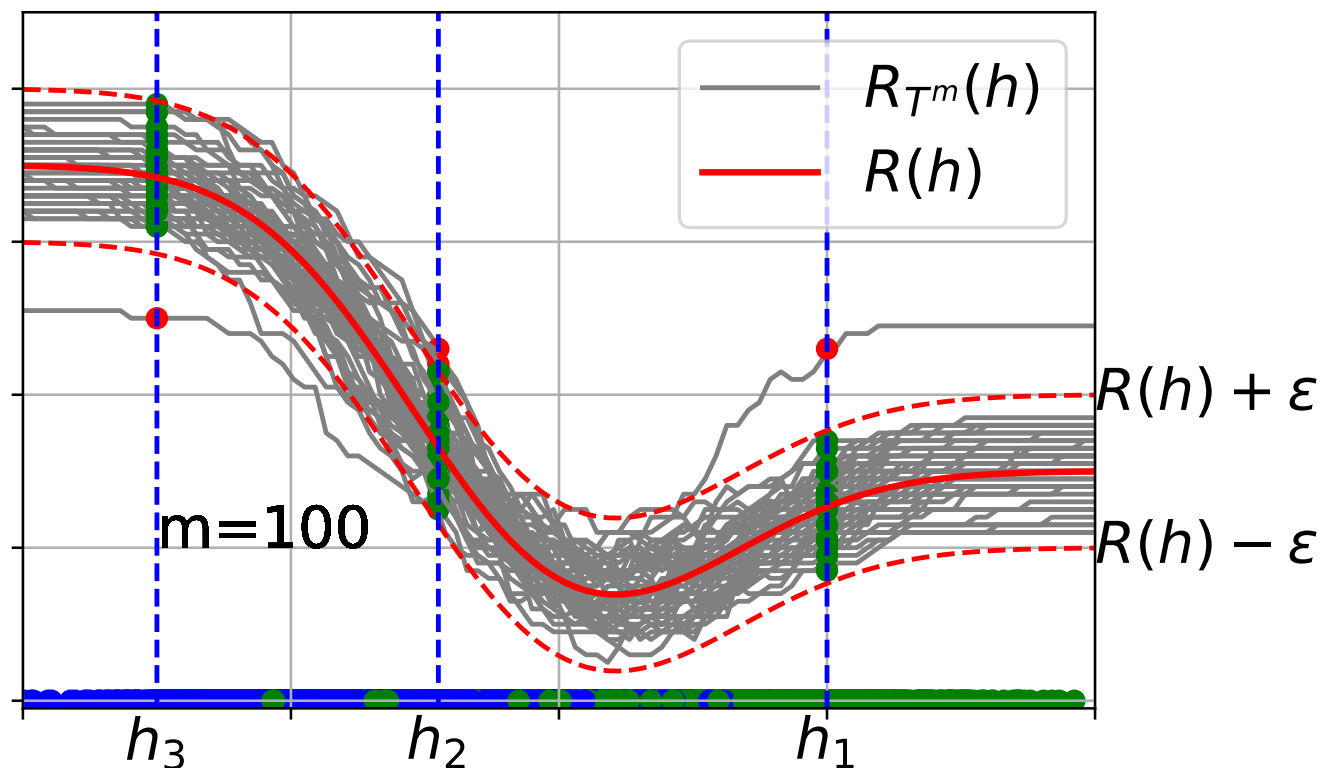
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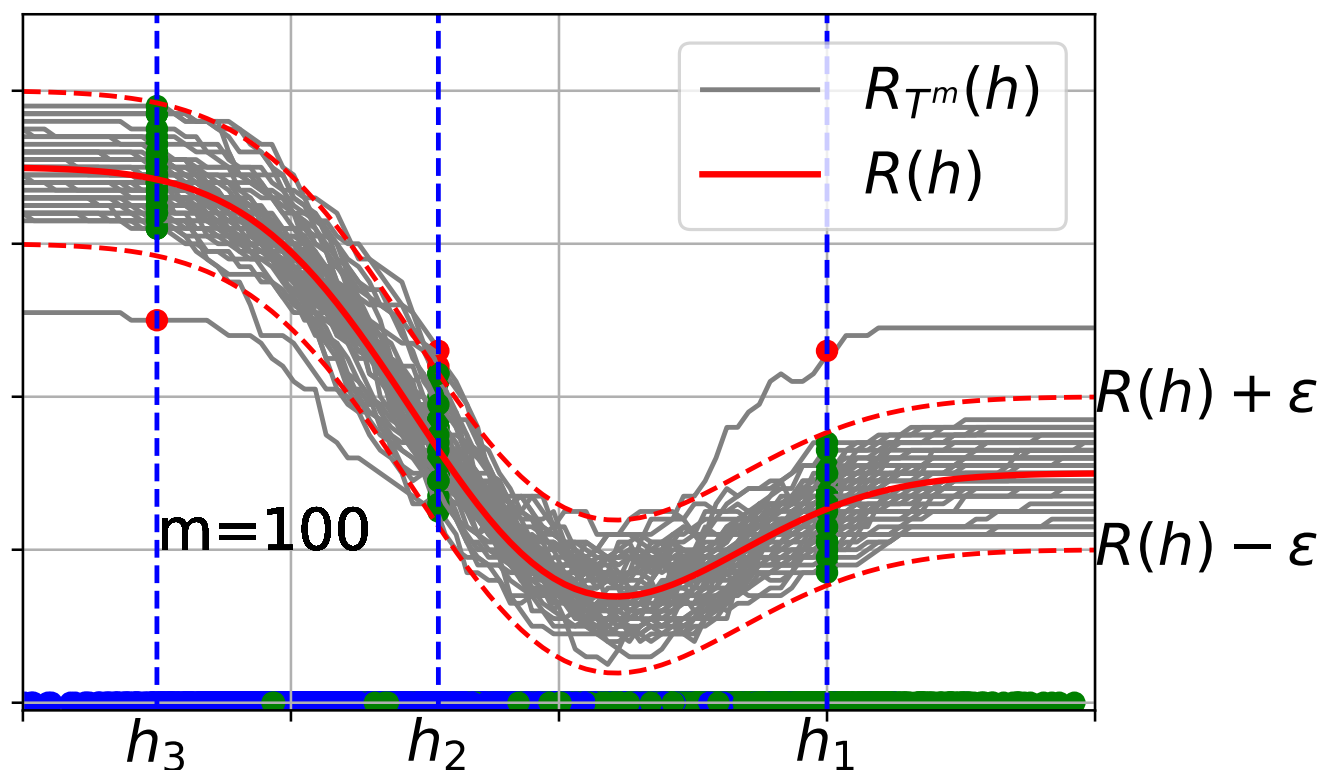


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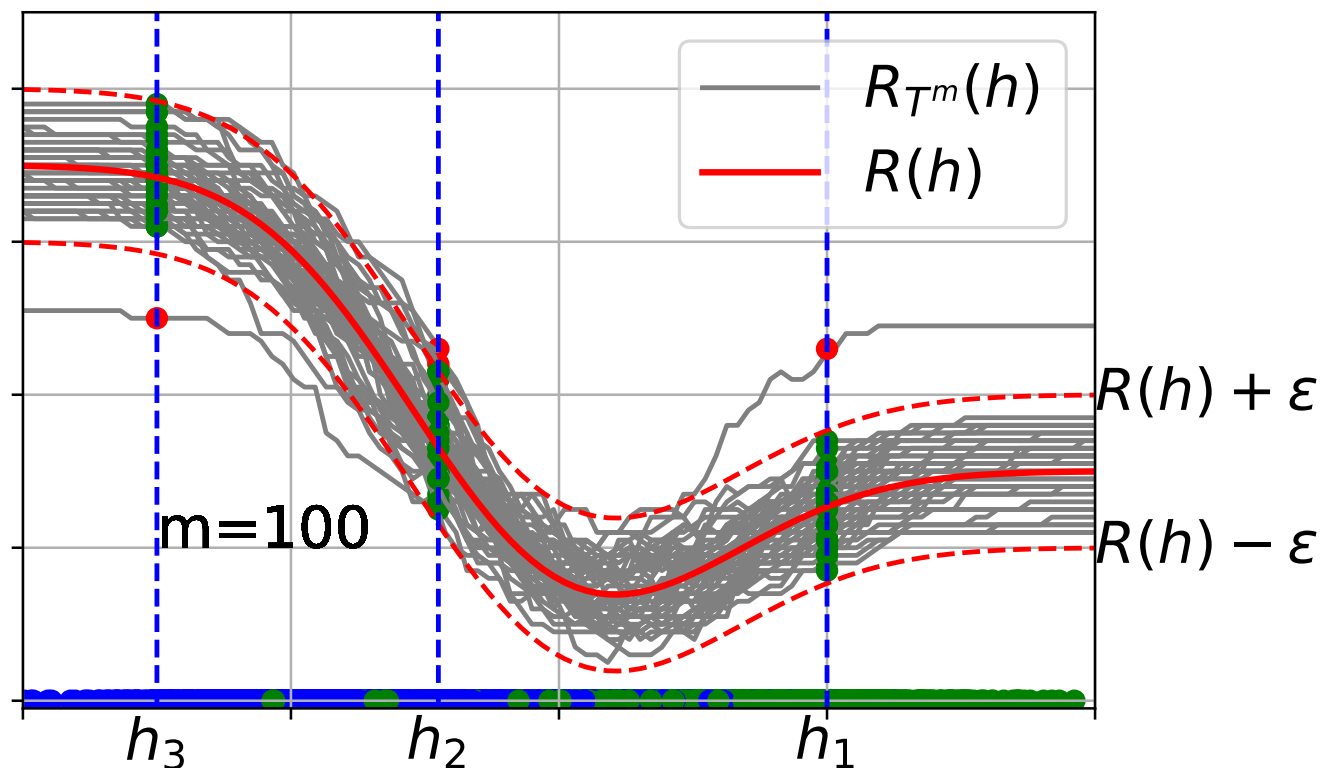


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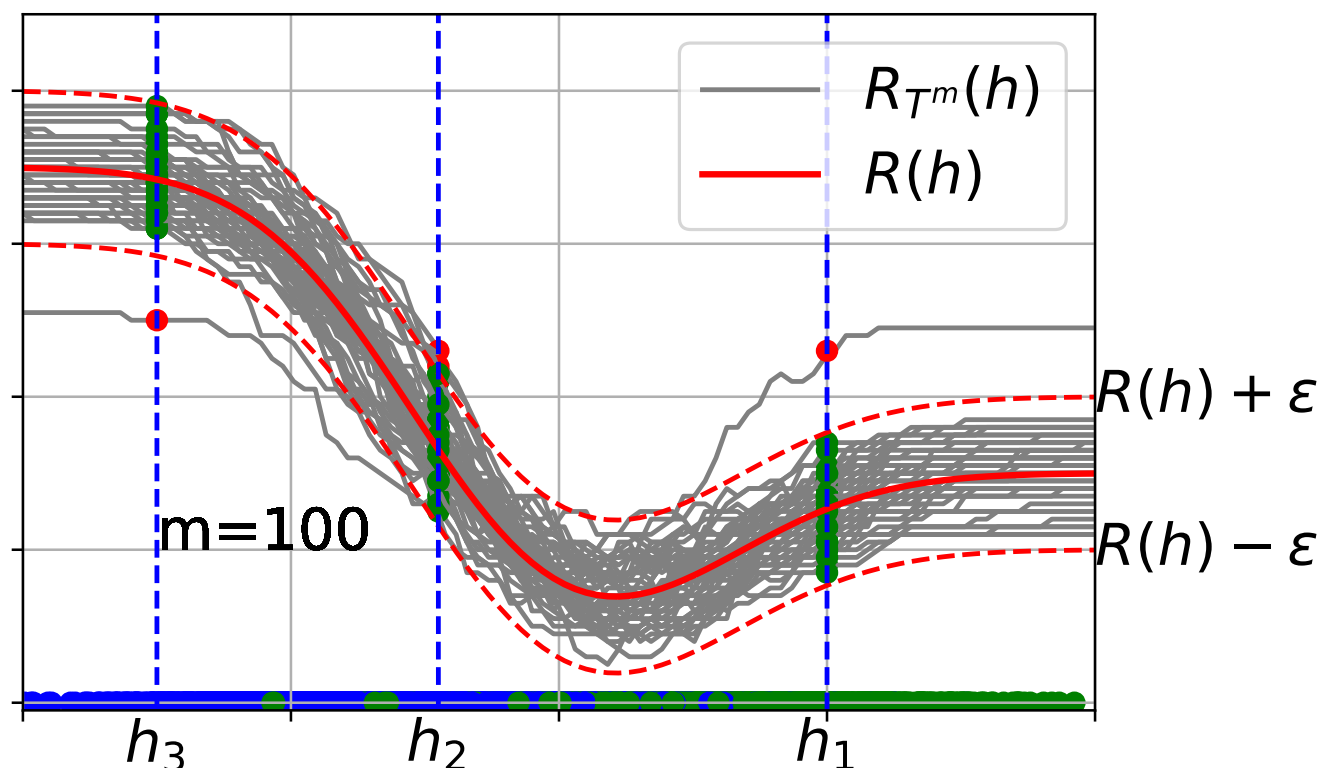
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# Uniform Law of Large Numbers

- ◆ **Definition:** We say that **Uniform Law of Large Numbers** applies for hypothesis space  $\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$  if there exists a function  $m_{\mathcal{H}}: \mathbb{R}_{>0} \times (0, 1) \rightarrow \mathbb{N}$ , called sample complexity, such that for every  $\varepsilon > 0, \delta \in (0, 1)$ , every distribution  $p(x, y)$  and every  $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$  the following inequality holds

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- ◆ **The issue:** How to recognize whether ULLN holds for a given hypothesis space  $\mathcal{H}$  ?
- ◆ We are going to show that ULLN applies for a finite hypothesis space  $\mathcal{H} = \{h_1, \dots, h_K\}$ .

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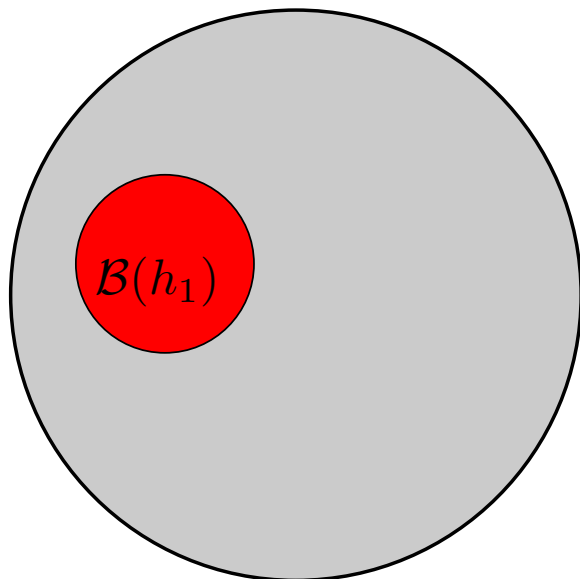
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$$a \geq \varepsilon \quad \text{or} \quad b \geq \varepsilon \quad \Longleftrightarrow \quad \max\{a, b\} \geq \varepsilon$$

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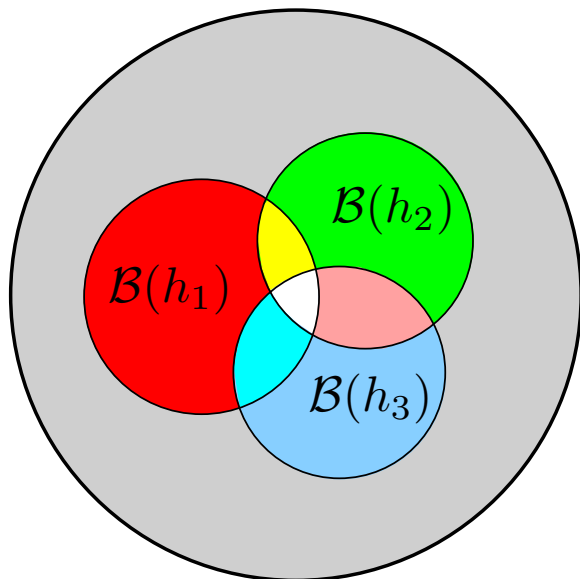


$$\mathcal{B}(h) = \{\mathcal{T}^m \in (\mathcal{X} \times \mathcal{Y})^m \mid |R_{\mathcal{T}^m}(h) - R(h)| \geq \varepsilon\}$$

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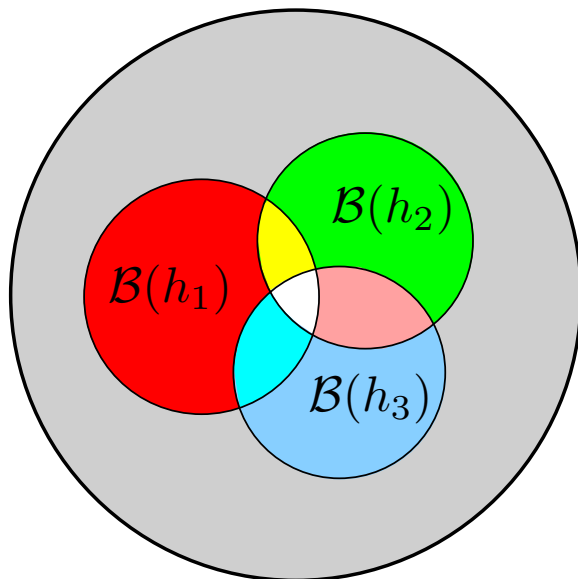


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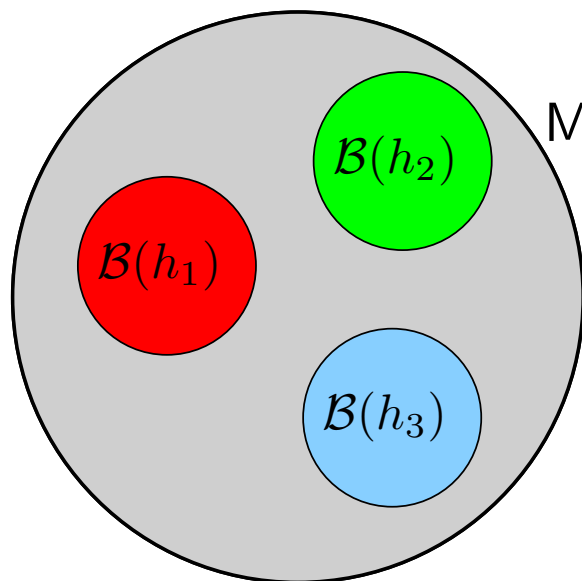


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Mutually exclusive events:  $\mathcal{B}(h_1) \cap \mathcal{B}(h_2) \cap \mathcal{B}(h_3) = \emptyset$

$$\mathbb{P}(\mathcal{B}(h_1) \cup \mathcal{B}(h_2) \cup \mathcal{B}(h_3)) =$$

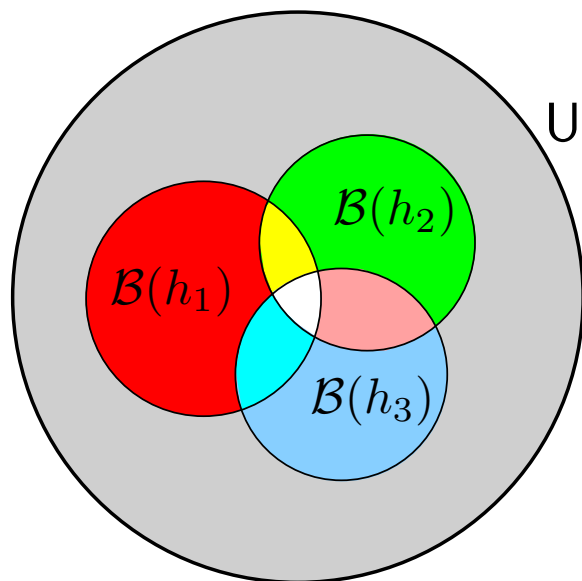
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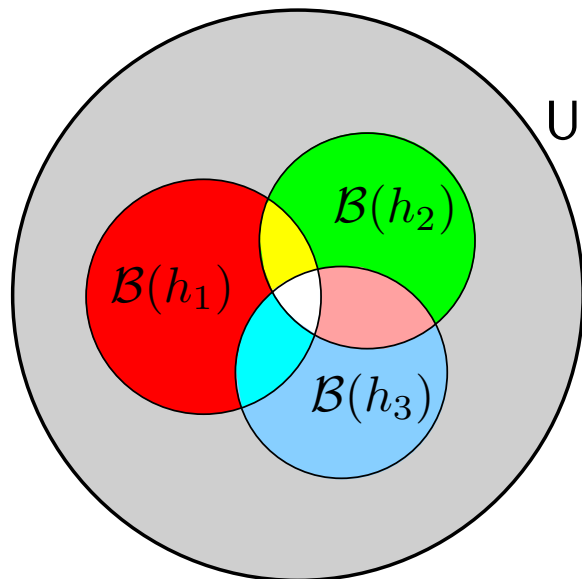
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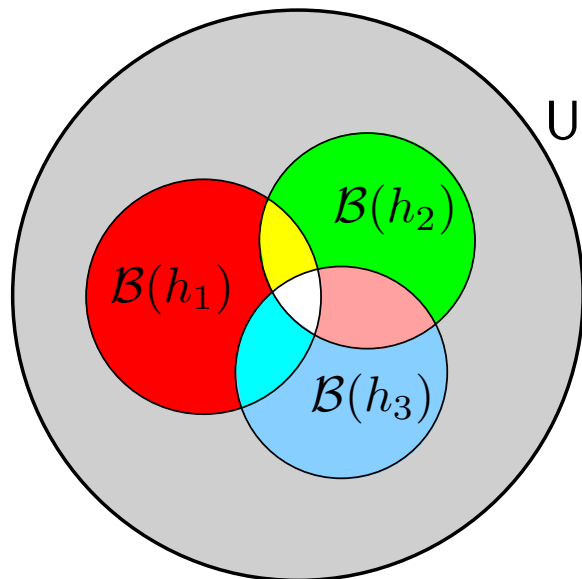
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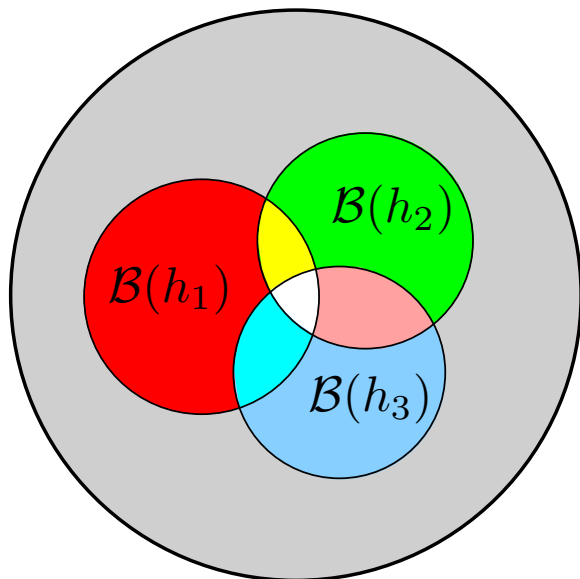
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Hoeffding inequality:

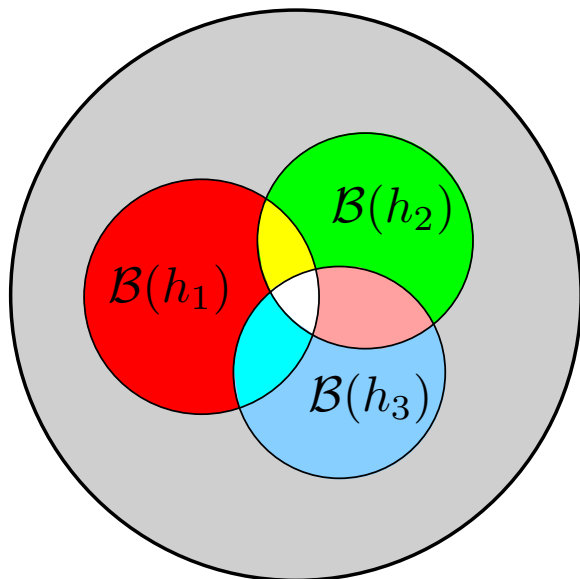
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- ◆ Assume a finite hypothesis class  $\mathcal{H} = \{h_1, \dots, h_K\}$ .

$$\begin{aligned} \mathbb{P}\left(\underbrace{\max_{h \in \mathcal{H}} |R(h) - R_{\mathcal{T}^m}(h)| \geq \varepsilon}_{\text{ER fails for some } h \in \mathcal{H}}\right) &\stackrel{(1)}{=} \mathbb{P}\left(\begin{array}{c} |R(h_1) - R_{\mathcal{T}^m}(h_1)| \geq \varepsilon \quad \text{or} \\ |R(h_2) - R_{\mathcal{T}^m}(h_2)| \geq \varepsilon \quad \text{or} \\ \vdots \\ |R(h_K) - R_{\mathcal{T}^m}(h_K)| \geq \varepsilon \end{array}\right) \\ &\stackrel{(2)}{\leq} \sum_{h \in \mathcal{H}} \mathbb{P}\left(|R(h) - R_{\mathcal{T}^m}(h)| \geq \varepsilon\right) \\ &\stackrel{(3)}{\leq} 2 |\mathcal{H}| e^{-\frac{2m\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}} \end{aligned}$$

1.  $a \geq \varepsilon \quad \text{or} \quad b \geq \varepsilon \iff \max\{a, b\} \geq \varepsilon$
2. Union bound.
3. Hoeffding inequality.

# Uniform Law of Large Numbers applies to the finite hypothesis space

- ◆ **Definition:** We say that **Uniform Law of Large Numbers** applies for hypothesis space  $\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$  if there exists a function  $m_{\mathcal{H}}: \mathbb{R}_{>0} \times (0, 1) \rightarrow \mathbb{N}$ , called sample complexity, such that for every  $\varepsilon > 0, \delta \in (0, 1)$ , every distribution  $p(x, y)$  and every  $m \geq m_{\mathcal{H}}^{\text{ul}}(\varepsilon, \delta)$  the following inequality holds

$$\mathbb{P}\left(\max_{h \in \mathcal{H}} |R(h) - R_{\mathcal{T}^m}(h)| \geq \varepsilon\right) \leq \delta.$$

- ◆ We have shown that for the finite hypothesis space,  $\mathcal{H} = \{h_1, \dots, h_K\}$ :

$$\mathbb{P}\left(\max_{h \in \mathcal{H}} |R(h) - R_{\mathcal{T}^m}(h)| \geq \varepsilon\right) \leq 2|\mathcal{H}|e^{-\frac{2m\varepsilon^2}{(\ell_{\max} - \ell_{\min})^2}}$$

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- ◆ **Corrolary:** For any loss  $\ell: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$  and a finite hypotheses space  $\mathcal{H} = \{h_1, \dots, h_K\}$ , the ULLN applies with the sample complexity:

$$m_{\mathcal{H}}^{\text{ul}}(\varepsilon, \delta) = \frac{\log(2|\mathcal{H}|) - \log \delta}{2\varepsilon^2} (\ell_{\max} - \ell_{\min})^2$$

# Generalization bound for finite hypothesis space

**Theorem:** Let  $\mathcal{T}^m = ((x^1, y^1), \dots, (x^m, y^m)) \in (\mathcal{X} \times \mathcal{Y})^m$  be draw from i.i.d. rv. with p.d.f.  $p(x, y)$  and let  $\mathcal{H}$  be a finite hypothesis space. Then, for any  $0 < \delta < 1$ , with probability at least  $1 - \delta$  the inequality

$$R(h) \leq \underbrace{R_{\mathcal{T}^m}(h)}_{\text{empirical risk}} + \underbrace{(\ell_{\max} - \ell_{\min}) \sqrt{\frac{\log 2|\mathcal{H}| + \log \frac{1}{\delta}}{2m}}}_{\text{complexity term}}$$

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- ◆ The generalization bound holds for any learning algorithm not just ERM.
- ◆ To decreases the complexity term: increase  $m$  or decrease  $|\mathcal{H}|$ .
- ◆ Recommendations for learning:
  1. Minimize the empirical risk.
  2. Use as much training examples  $m$  as you can.
  3. Limit the size of the hypothesis space  $|\mathcal{H}|$ , i.e. use prior knowledge.

## Structural Risk Minimization

- ◆ Learn  $h: \mathcal{X} \rightarrow \mathcal{Y}$  by minimizing the generalization bound

$$R(h) \leq R_{\mathcal{T}^m}(h) + \underbrace{(b-a) \sqrt{\frac{\log 2|\mathcal{H}| + \log \frac{1}{\delta}}{2m}}}_{\epsilon(m, |\mathcal{H}|, \delta)}$$

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- ◆ Design a nested sequence of hypothesis classes

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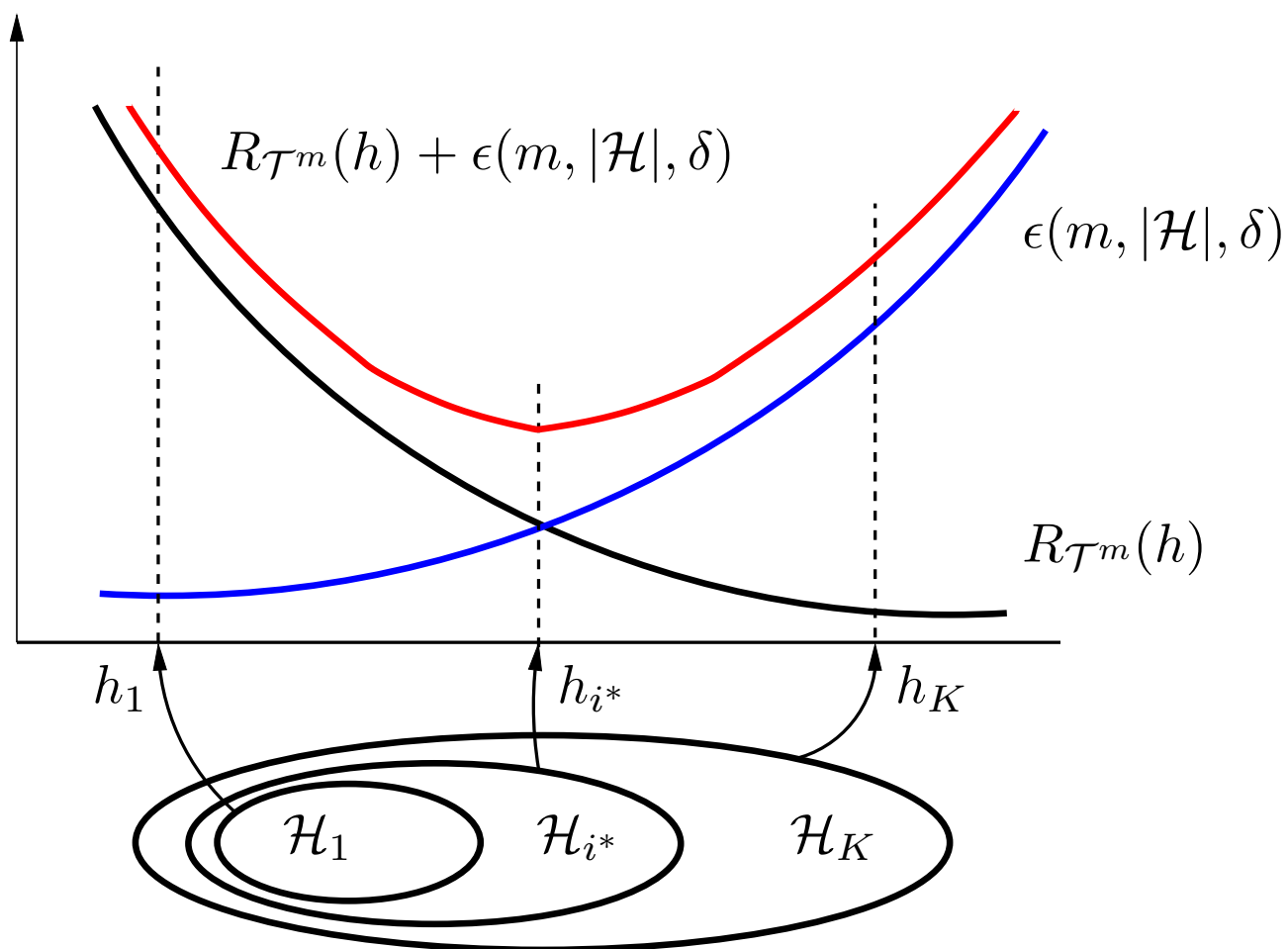
- Minimize the generalization bound:

- $h_i = \operatorname{argmin}_{h \in \mathcal{H}_i} R_{\mathcal{T}^m}(h), \quad \forall i \in \{1, \dots, K\}$
- $i^* = \operatorname{argmin}_{i=1, \dots, K} \left( R_{\mathcal{T}^m}(h_i) + \epsilon(m, |\mathcal{H}_i|, \delta) \right)$
- Output  $h_{i^*}$

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## Summary

- ◆ Empirical Risk is not in general a reliable proxy of the expected risk.
- ◆ For ERM based learning we need the Uniform Law of Large Numbers (ULLN).
- ◆ Whether ULLN applies or not depends on the particular  $\mathcal{H}$ .
- ◆ ULLN applies for the finite hypothesis space.
- ◆ Generalization bound for finite hypothesis space.
- ◆ Structural Risk Minimization.

